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SUPERSEDING
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MILITARY SPECIFICATION

PROPELLANT M30 FOR USE IN CARTRIDGE 105MM, APFSDS-T, M833

This specification is approved for use by the US Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers M30 Propellant for use in the Cartridge APFSDS-T, M833 for the 105MM, M68 Gun.

1.2 Classification. The propellant shall be 19-perforated, hexagonal grains.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 -Ammunition, Standard Quality Assurance Provisions, General Specification for

FSC 1376

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn. DRSMC-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-286	-Propellants, Solid: Sampling, Examination And Testing
MIL-STD-652	-Propellants, Solid, for Cannons, Requirements and Packing

2.1.2 Other government documents, drawings and publications. The following other Government documents form a part of this specification to the extent specified herein.

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER

9296617	-Primer, Electric, M120, Loading Assembly
9342932	-Cartridge, 105MM, APFSDS-T, M833
9370131	-19 Perforated Rounded Hexagonal Configuration
10522799	-Case Cartridge, 105MM, M148A1B1

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Material. The propellant shall be M30, with a nominal web of 0.050 inch conforming to the requirements of MIL-STD-652 and applicable cartridge drawing except for the following:

3.1.1 Composition

3.1.1.1 Total volatiles. The propellant lot total volatile content shall be 0.25 percent maximum and the composite sample shall be between 0.1 to 0.30 percent.

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3.1.1.2 Graphite content. The graphite (glaze) content shall not exceed 0.2 percent as determined in accordance with 4.5.4.

3.1.1.3 Closed bomb. Closed bomb testing shall be in accordance with 4.5.5 for information.

3.1.1.4 Heat of explosion (HOE). If the HOE calories per gram (cal/gm) from the determined composition varies by more than 7 cal/gm from the measured HOE, then tests shall be repeated and all data reported to the technical agency. Acceptance of the lot will be determined on the subsequent chemical analysis.

3.1.2 Type. The grain shall be rounded hexagonal with 19 circular longitudinal perforations, one in the center of the grain, six at the vertices of an inner symmetrical hexagon, and twelve forming an outer symmetrical hexagon as is shown in figure 1 on drawing 9370131. The grains when tested in accordance with 4.4.2.1.2.1 and 4.5.3 shall meet the criteria of Table I. Dimensions listed in Table I as average are defined as the average value of 40 grains.

TABLE I. Physical properties

Average grain length (L) to average large grain diameter (D_L) ratio (L/D_L)	1.25 ± 0.1
Average large grain diameter (D_L) to average perforation diameter (d) ratio (D_L/d)	18.0 ± 2.0
Percent web difference	15 percent maximum
Perforation	See 3.1.2.1
Bulk density	63 lb/ft^3 minimum
Grain physical appearance score	100 minimum

3.1.2.1 Perforation. The perforation diameter standard deviation shall be less than 0.0025 inch and there shall not be any missing perforations.

3.1.2.2 Grain dimensions (measurement of grain ends). Measurements of grain ends shall be performed in accordance with 4.4.2.1.2.2 on the first three production lots, any selected pilot lot, and the subsequent production lot.

3.2 Storage. Packaged propellant shall be stored a minimum of twenty-one days from final packout to uniformity testing.

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3.3 Ballistic assessment requirements. The propellant when fired in the M68 Gun with the projectile M833 shall comply with the following requirements at 70 degrees Fahrenheit (^{O}F) and standard weapon conditions. The propellant shall be capable of assessment to a velocity of 4900 feet per second (fps) with an associated pressure not greater than 67000 pounds per square inch (psi) (see 6.7).

3.3.1 Maximum charge weight. The recommended charge weight shall be no greater than 215.0 ounces.

3.4 Ballistic uniformity requirements. The propellant when fired in the M68 Gun with the projectile M833 shall comply with the following requirements as determined in accordance with 4.5.2.2 (see 6.7).

3.4.1 Uniformity of velocity. The standard deviation of velocity at 70°F shall not exceed 9.6 fps.

3.4.2 Average pressure. At 70°F and standard weapon conditions, the propellant shall not produce an average pressure greater than 67000 psi.

3.4.3 Individual maximum pressure (IMP). At 125°F and standard weapon conditions the propellant shall not produce an individual maximum pressure greater than 75,700 psi nor shall the average pressure plus three standard deviations exceed 75,700 psi.

3.4.4 Negative differential pressure. Differential pressures shall be measured in accordance with 4.5.2.2.4 for information.

3.5 Preproduction inspection. This specification contains provisions for preproduction inspection. Requirements for the submission of preproduction inspection samples by the contractor shall be as specified in 4.3, 4.4.2.1.2 and 4.5.2.

3.6 Initial production. This specification contains provisions for initial production inspection. Requirements for the submission of initial production inspection samples by the contractor shall be as specified in 4.5.2.

3.7 Workmanship. The best commercial practices shall be used in the formulation of propellant furnished under this specification, and all other applicable documents. The propellant and its standard ingredients shall be protected from the action of direct sunlight and acid fumes. Unless otherwise specified, no reground propellant or nitrocellulose shall be used in the manufacture of propellant procured under this specification. The amount of remix propellant added to each mixer shall not deviate by more than 10 percent from the established amount of remix propellant of all mixes comprising the propellant lot.

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4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on this unit of product:

a. Preproduction Sample Inspection
b. Quality Conformance Inspection

4.3 Preproduction sample (pilot lots). The preproduction sample is intended to provide the necessary information to establish the propellant granulation needed to meet the required ballistic parameters.

4.3.1 Submission. The Contractor shall select propellant granulations for approval by Government Engineering and Product Assurance, which, from experience, are expected to bracket the requirements of this specification. The contractor shall submit a preproduction sample as designated by the Contracting Officer for evaluation in accordance with 4.3.2, usually consisting of up to three pilot lots of approximately 2500 lb each, representing each of the candidate granulations.

4.3.2 Inspections to be performed. Each of the pilot lots comprising the preproduction sample and submitted in accordance with 4.3.1 may be subjected by the Government to any or all of the requirements of the applicable drawings and specifications.

4.3.3 Rejection. Government Engineering and Product Assurance personnel and the Contractor shall review the pilot lot test data and shall select the granulation which best fits the requirements of this specification. The Technical Agency shall notify the Contracting Officer of the selected granulation. The Contracting Officer shall designate the chosen pilot lot as the accepted preproduction sample and shall authorize the Contractor to produce the selected granulation. If none of the pilot lots submitted meets or brackets the requirements of this specification, a new preproduction sample shall be submitted in accordance with 4.3.1. Until the preproduction sample is accepted, the Contractor is in no way authorized by the Government to initiate regular production, unless otherwise directed by the Contracting Officer. All future production lots should be of the same basic granulation as the accepted pilot lot.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination and testing.

PRECAUTION: This specification covers sampling and testing of chemical, toxic or explosive materials which are potentially hazardous to personnel. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing these materials.

4.4.2.1 Sampling for laboratory testing (see 3.1). If any samples fail to meet the applicable requirement, the lot shall be rejected. Each lot of propellant shall be sampled and tested for composition, form, dimensions and chemical/physical properties in accordance with MIL-STD-652, except for the following:

4.4.2.1.1 Samples for grain physical appearance score. A sample of 100 randomly selected grains shall be visually inspected from each inspection sublot of propellant prior to drying. An inspection sublot shall be no greater than 10 cabinets. The number of grains sampled from each cabinet shall be approximately equal.

4.4.2.1.1.1 Defect criteria. The grains will be visually inspected for the following defects. The number of defects (T) found for each category will be recorded.

4.4.2.1.1.1.1 Category I (surface). A defect that will appreciably increase the surface area of the grain, as follows:

4.4.2.1.1.1.1.1 Grains marked by die inperfections causing longitudinal breaks in the surface of the grain.

4.4.2.1.1.1.1.2 Mashed or crushed grains in which the perforation is closed or the grain is cracked.

4.4.2.1.1.1.1.3 Blisters or enclosed raised spots on the grain.

4.4.2.1.1.1.1.4 Foreign material, e.g., wood, metal specks, brush hairs, grease, etc.

4.4.2.1.1.1.1.5 Cutting machine, roller indentations pushed into the grain surface.

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4.4.2.1.1.1.1.6 Barbed surface, typically slivers or blobs will protrude from the grain surface.

4.4.2.1.1.1.2 Category II (perforation). A defect caused by improper location of the perforation, as follows:

4.4.2.1.1.1.2.1 Closed perforation. Grains with any one of the perforations closed.

4.4.2.1.1.1.2.2 Partially closed perforations. Grains with any one of the perforations appreciably different from the others (perforations in a grain must not vary in size by more than a 1:.75 ratio).

4.4.2.1.1.1.2.3 Off-center perforations or distorted web. Grains in which the distance between any 2 adjacent perforations is approximately 1/3 less than the distance between any other 2 adjacent perforations or in which the distance from an outer perforation to the surface of the grain is 1/3 less than the similar distance from another outer perfortion.

4.4.2.1.1.1.3 Category III (cracks). The surface of a grain shall have no breaks, rupture, or cuts.

4.4.2.1.1.1.4 Category IV (pinholes). Grains shall have no visible small holes in the surface (use of a magnifying glass is not permitted).

4.4.2.1.1.1.5 Category V (bad cuts). A defect caused by cutting machine action, as follows:

4.4.2.1.1.1.5.1 Grains shorter than approximately 3/4 of the specified length of propellant being inspected.

4.4.2.1.1.1.5.2 Grains which contain cuts in the grain body made by machine blades.

4.4.2.1.1.2 Calculation of score. the score shall be calculated as folows:

$$\text{Score} = 200 - \left(T_I + 2T_{II} + 6T_{III} + 4T_{IV} + 2T_V \right)$$

where

T_I = No. of category I defects observed
T_{II} = No. of category II defects observed.
T_{III} = No. of category III defects observed.
T_{IV} = No. of category IV defects observed.
T_V = No. of category V defects observed.

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4.4.2.1.1.3 Rejection. If the grain physical appearance score fails to meet the requirement the inspection sublot represented by the inspected sample shall be rejected.

4.4.2.1.2 Sampling for grain dimensions shall be as follows:

4.4.2.1.2.1 Internal measurements. All lots, including pilot lots, shall have 40 grains randomly selected from the lot. Testing will be in accordance with 4.5.1.1.

4.4.2.1.2.2 End measurements.

4.4.2.1.2.2.1 Production sampling. The first three lots manufactured shall be inspected. Upon completion of a lot, a 100 grains shall be randomly sampled and measured on the ends in accordance with 4.5.1.2.

4.4.2.1.2.2.2 Pilot lot sampling. The pilot lot closest to the selected granulation and the first production lot shall have 30 grains randomly selected and measured on the ends in accordance with 4.5.1.2.

4.4.2.2 Sampling for ballistic testing.- Each lot of propellant shall be sampled for Proving Ground tests in accordance with MIL-STD-652. If the lot fails to meet any of the following criteria, apply retest and referral criteria in accordance with 4.5.2.3 and 4.5.2.4. If these criteria do not apply the lot shall be rejected.

4.4.2.3 Charge establishment (see 3.3).- If the lot does not meet the applicable requirements, testing shall cease and the Test Director shall immediately notify Commander, Armament Research and Development Center, US Army Armament, Munitions and Chemical Command, ATTN: DRSMC-QAR-R (D), Dover, NJ 07801 and Commander, US Army Armament, Munitions and Chemical Command, ATTN: DRSMC-QAM-A (R), Rock Island, IL 61299.

4.4.2.4 Velocity uniformity (70°F) (see 3.4).- If the product of the velocity standard deviation of the rounds tested in accordance with 4.5.2.2 times the factor 0.73 exceeds the applicable requirement, the lot shall be rejected.

4.4.2.5 Average pressure (see 3.4). If the average pressure obtained exceeds the applicable requirement, the lot shall be rejected.

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4.4.2.6 Individual maximum pressure (see 3.4).

4.4.2.6.1 If any individual pressure observed during the uniformity series, when corrected to standard conditions and recommended charge weight exceeds the applicable requirement, the lot shall be rejected.

4.4.2.6.2 If the average pressure plus three standard deviations exceeds the applicable requirement, the lot shall be rejected.

4.5 Test methods and procedures.

4.5.1 Manufacturing assessment.

4.5.1.1 Grain dimensions (internal measurements).

4.5.1.1.1 Each grain shall be partially cut at the center of the grain and parallel to the end, and broken apart by hand to obtain the measurements shown in Figure 2.

4.5.1.1.2 Each grain shall be measured for length (L) large grain diameter (D_L), small grain diameter (D_S), perforation diameters (d), outer webs (Wo), middle webs (Wm), inner webs (Wi), around- outer webs (Wao), and Z. All dimensions shall be measured with an image analyzer, except the grain length and grain diameter may be measured with a micrometer, as shown on Figure 2 and as described in the following:

4.5.1.1.2.1 Length (L). The length shall be measured at the average length location.

4.5.1.1.2.2 Grain diameter (D_L) and (D_S). The large diameter (D_L) and small diameter (D_S) shall be measured.

4.5.1.1.2.3 Perforation diameter (d). Seven perforations (d_1, d_2,...d_7) shall be measured.

4.5.1.1.2.4 Web outer (Wo). Four outer webs (Wo_1, Wo_2,...Wo_4) shall be measured.

4.5.1.1.2.5 Web middle (Wm). Four middle webs (Wm_1. Wm_2,...Wm_4) shall be measured.

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4.5.1.1.2.6 Web inner (Wi). Two inner webs (Wi_1 and Wi_2) shall be measured.

4.5.1.1.2.7 Web around-outer (Wao). Two around-outer webs (Wao_1 and Wao_2) shall be measured.

4.5.1.1.2.8 Z. Two Z dimensions (Z_1 and Z_2) shall be measured.

4.5.1.1.3 Grain calculations. Based on a 40 grain random sample calculate a lot average value and a standard deviation for Wo, Wm, Wi, Wao, Z, and d. Also calculate the weighted average web ($\overline{W}$) as follows:

$$\overline{W} = \frac{2\,\overline{W}o + 2\,\overline{W}m + \overline{W}i + \overline{W}ao}{6}$$

$\overline{W}o$ = lot average outer web
$\overline{W}m$ = lot average middle web
$\overline{W}i$ = lot average inner web
$\overline{W}ao$ = lot average around-outer web

4.5.1.1.4. Percent web difference calculation. Calculate the percent web difference as follows:

$$\% \text{ web difference} = 100 \times \frac{|\overline{W}o - \overline{W}m| + |\overline{W}m - \overline{W}i| + |\overline{W}o - \overline{W}i|}{3 \times \overline{W}}$$

$\overline{W}o$ = lot average outer web
$\overline{W}m$ = lot average middle web
$\overline{W}i$ = lot average inner web
$\overline{W}$ = lot weighted average web

4.5.1.2 Grain dimensions (end measurements).

4.5.1.2.1 Each grain shall be measured for lengths (L), large grain diameters (D_L), small grain diameters (D_S), perforation diameters (d), outer webs (Wo), middle webs (Wm), inner webs (Wi), around-outer webs (Wao), and around-inner webs (Wai). Each dimension shall be recorded on the same end of the grain as shown on drawing 9370131 with its relationship traceable to the perforation selected as d_1. All dimensions shall be measured with an image analyzer, except the grain length and grain diameters may be measured with a micrometer, as described in the following:

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4.5.1.2.1.1 Length (L). The length shall be measured at the largest (L_L) and smallest (L_S) dimensions.

4.5.1.2.1.2 Grain diameter (D_L) and (D_S). The large diameters (D_{L1}, D_{L2}, and D_{L3}) and the small diameters (D_{S1}, D_{S2}, and D_{S3}) shall both be measured as shown in Figure 2 on drawing 9370131.

4.5.1.2.1.3 Perforation diameter (d). Nineteen perforations (d_1, d_2.....d_{19}) shall be measured on the grain as shown in Figure 3 on drawing 9370131.

4.5.1.2.1.4 Web outer (Wo). Twelve outer webs (Wo_1, Wo_2, ... Wo_{12}) shall be measured on the grain as shown in Figure 4 on drawing 9370131.

4.5.1.2.1.5 Web middle (Wm). Eighteen middle webs (Wm_1, Wm_2, ... Wm_{18}) shall be measured on the grain as shown in Figure 5 on drawing 9370131.

4.5.1.2.1.6 Web inner (Wi). Six inner webs (Wi_1, Wi_2, ... Wi_6) shall be measured on the grain as shown in Figure 6 on drawing 9370131.

4.5.1.2.1.7 Web around-outer (Wao). Twelve around-outer webs (Wao_1, Wao_2, ... Wao_{12}) shall be measured on the grain as shown in Figure 7 on drawing 9370131.

4.5.1.2.1.8 Web around-inner (Wai). Six around-inner webs (Wai_1, Wai_2, ... Wai_6) shall be measured on the grain as shown in Figure 8 on drawing 9370131.

4.5.1.2.2 Data. All data shall be recorded using the data sheet, Figure 1, and submitted to Commander, Armament Research and Development Center, US Army Armament, Munitions and Chemical Command, ATTN: DRSMC-QAR-R (D), Dover, NJ 07801 and Commander, US Army Armament, Munitions and Chemical Command, ATTN: DRSMC-QAM-A (R), Rock Island, IL 61299.

4.5.2 Proving ground assessment. These tests shall be conducted at a Government proving ground (designated by the Contracting Officer) in accordance with the applicable Proving Ground Acceptance Test Procedure. Tubes used for these assessments shall be restricted to firing only Armor Piercing Fin Stabilizing Discarding Sabot with Tracer (APFSDS-T)

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ammunition. The wear of the tube shall be no greater than 0.015 in. when measured 25.25 in. from the rear face of the tube. All test rounds shall be temperature conditioned in a large conditioning box, with adequate air circulation to assure uniform conditioning, at the required temperature for a minimum of 24 hours and assurance will be made, when fired, that the rounds are at the conditioning temperature as indicated by thermocoupled control rounds, 2 per conditioning unit. The following tolerances shall apply to all temperature conditionings: $-35^{\circ}F + 5^{\circ}F$; $70^{\circ}F \pm 2.5^{\circ}F$; $125^{\circ}F - 5^{\circ}F$. The as-fired projectile weight shall be 13.45 $\pm$ 0.05 lb and all data shall be corrected to a projectile weight of 13.45 lbs. Copper crusher gages (2) shall be used to measure the maximum chamber pressure. Test rounds shall be assembled in accordance with the applicable drawing. Three skyscreens will be used to measure the muzzle velocity. The difference between skyscreen readings ((closest to middle) minus (closest to furthest) and (closest to furthest) minus (middle to furthest)) shall not be a negative number (reversal) or exceed 15 fps (closest, middle, and furthest represents distance from weapon). The distance between skyscreens shall not exceed 100 feet. The cartridges (reference, calibration and test) shall be loaded and inspected by the Army Ammunition Plant LAP facility or similar manufacturer of cartridge lots in accordance with drawing 9342932 and shipped to the proving ground for ballistic testing.

4.5.2.1 Charge establishment.

4.5.2.1.1 Charge slope method. Three or more groups of rounds loaded with selected charge weights of propellant (weights selected should bracket expected service charge weight) shall be conditioned and fired at $70^{\circ}F$. Three calibration rounds shall also be conditioned and fired at $70^{\circ}F$, prior to the test rounds. From the data obtained, a charge weight shall be calculated which will yield the prescribed service velocity (see 3.3). This calculated charge weight shall then be loaded into rounds for the uniformity series (see 4.5.2.2.). The difference between the corrected velocity during the uniformity series and the service velocity shall be compensated for by a final adjustment in charge weight using the slope of the velocity versus charge weight curve . The resultant will be the recommended charge weight.

4.5.2.1.2 Constant slope method. After ten lots of a given production series have been fired to provide a good knowledge of the propellant characteristics, the charge slope method of assessment may be discontinued and the charge weight can be determined by the constant slope method. A composite charge weight versus velocity curve can be established from test results

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generated. This curve can then be used as in 4.5.2.1.1 to calculate the charge weight needed to achieve service velocity. This calculated charge weight is then to be loaded into rounds for the uniformity series and final adjustment is to be made as indicated above for the charge slope method (see 4.5.2.1.1). The resultant will be the recommended charge weight.

4.5.2.2 Uniformity. Firing results at all temperatures shall be corrected for the presence of M11 gages, to standard projectile weight (see 4.5.2), and to recommended charge weight conditions (i.e., the difference between as fired charge weight and recommended charge weight). The means and standard deviations of the velocities and pressures at all firing temperatures shall be calculated using corrected results. The initial production lots represent the first five consecutive lots from each manufacturer which have been produced and met the acceptance criteria of this specification. Testing shall revert back to initial production if production has stopped for more than one year.

4.5.2.2.1 Preproduction. Thirty rounds of each pilot lot shall be assembled with the calculated charge weight (see 4.5.2.1.1) of propellant. Ten rounds each shall be conditioned to -35°F, 70°F, and 125°F (as indicated by thermocoupled control rounds, 2 per conditioning unit). Rounds fired at 70°F shall be fired alternately with calibration rounds. Rounds fired at -35°F and 125°F will be fired with five (5) calibration rounds intermixed at each temperature.

4.5.2.2.2 Initial production. Thirty rounds shall be assembled with the calculated charge weight (see 4.5.2.1.1) of propellant. Ten rounds each shall be conditioned to -35°F, 70°F, and 125°F(as indicated by thermocoupled control rounds, 2 per conditioning unit). Rounds fired at 70°F shall be fired alternately with calibration rounds and shall be corrected to standard weapon/ firing conditions (i.e., the difference between calibration performance under test conditions and standard conditions). Rounds fired at 125°F will be intermixed with five (5) calibration rounds. Differential pressure and ignition delay shall be obtained during initial production testing in accordance with 4.5.2.2.4.

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4.5.2.2.3 Subsequent production. Every fifth lot will be tested in accordance with 4.5.2.2.2. For the other four lots, twenty rounds shall be assembled with the calculated charge weight (see 4.5.2.1) of propellant. Ten rounds each shall be conditioned to 70°F and 125°F (as indicated by thermocoupled control rounds, 2 per conditioning unit). Rounds fired at 70°F shall be fired alternately with calibration rounds and shall be corrected to standard weapon/firing conditions (i.e., the difference between calibration performance under test conditions and standard conditions). Rounds fired at 125°F will be intermixed with five calibration rounds. Throughout production, if 3 consecutive lots fail on first test, testing shall revert back to the initial production plan.

4.5.2.2.4 Differential pressure and ignition delay (initial production). Two external pressure transducers mounted 14.25 (rear) inches and 20.5 (forward) inches from the rear face of the tube shall be used for measuring chamber pressure as a function of time during the -35°F and 125°F uniformity phases. Differential pressures shall be determined by subtracting the forward gage pressure from the respective rear gage pressure and recording the maximum initial negative difference. Any trace which does not contain any amplitude in the negative direction shall have a negative differential pressure of zero. Ignition delay shall be recorded on all differential pressure firings as the time required to obtain 10% of the maximum pressure on the rear gage from intiation of the primer.

4.5.2.3 Retest criteria. All original and retest data shall be submitted to Commander, Armament Research and Development Center, US Army Armament, Munitions and Chemical Command, ATTN: DRSMC-QAR-R (D), Dover, NJ 07801. A lot shall be retested whenever any of the following conditions exist:

a. The final adjustment in charge weight exceeds 2.0 oz of the charge weight used in uniformity firings retest for 70°F uniformity in accordance with 4.5.2.2.

b. If for any reason, the Proving Ground considers that test conditions have detrimentally affected test results, additional rounds as required shall be tested.

4.5.2.4 Referral criteria. A test lot shall be referred to the procuring activity for disposition whenever any of the following conditions exist or whenever the proving ground deems it necessary:

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a. Calibration rounds fired in any test phase fail to comply with Section 3.4 except 3.4.4 of this specification.

b. Calibration correction applied exceeds 25 fps.

c. Number of test or calibration results utilized in calculations is less than specified.

d. Any unusual occurrences during ballistic testing, such as excessive delays between rounds, poor projectile flights, etc. Each such event shall be reported in detail on the applicable firing record.

e. Test lot fails and any test value is an outlier. The following test shall be performed to determine if a single test result is an outlier. With a sample size of 10 at a significance level of 0.05, the critical value (T_c) is 0.477. Calculate a test value (T_t) as follows:

For Low Outlier	For High Outlier
$T_t = \frac{(X_2 - X_1)}{(X_9 - X_1)}$	$T_t = \frac{(X_{10} - X_9)}{(X_{10} - X_2)}$

X_1 or X_{10} is an outlier if T_t is greater than T_c.

Where:

X_1 = Test result being tested as low outlier.
X_2 = Test result adjacent to X_1, after all the test results are ordered from smallest to largest.
X_9 = Test result adjacent to the largest test result.
X_{10} = Test result being tested as high outlier.

4.5.3 Bulk density. The bulk density shall be determined in accordance with Method 502.2.1 of MIL-STD-286 except that a sample size of approximately 0.25 cubic feet shall be used and 1000 milliliter graduate cylinder shall be replaced with an M148A1B1 cartridge case, Dwg. 10522799, with M120 Primer, Dwg. 9296617.

4.5.4 Graphite (Glaze). The graphite content shall be determined in accordance with Method 308.1 of MIL-STD-286.

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4.5.5 Closed bomb. Testing and calculations shall be conducted on each composite sample and lot of propellant as described in MIL-STD-286 except for the following:

4.5.5.1 Standard loading density. Testing shall be conducted on samples conditioned to $-40^{\circ} + 5^{\circ}F$ and $90^{\circ} \pm 2.5^{\circ}F$ at a loading density of 0.2 gram per cubic centimeter (gm/cc) in a 700 cc vessel.

4.5.5.2 High loading density. Testing shall be conducted on samples conditioned to $70^{\circ} \pm 2.5^{\circ}F$ and $125^{\circ} - 5^{\circ}F$ at a loading density of approximately 0.30 gm/cc in a 700 cc vessel. Calculate a burning rate equation thru 60000 psi for both temperatures.

5. PACKAGING

5.1 Packing and marking. Packing and marking shall be in accordance with MIL-STD-652, except that all propellant designated for use as a reference shall be packed Level A (see 6.5).

6. NOTES

6.1 Ordering data. Ordering data shall be in accordance with MIL-A-48078 with the following exceptions:

6.1.1 Acquisition requirements. Acquisition requirements of MIL-A-48078 shall apply.

6.1.2 Contract data requirements. One copy of each of the Propellant Description Sheet and the Acceptance Test Summary shall be forwarded to each office designated by the Contracting Officer.

6.2 Proving ground test summary (see Table II).

6.3 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Center (ARDC) may also include drawings prepared by, and identified as, ARRADCOM, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDC.

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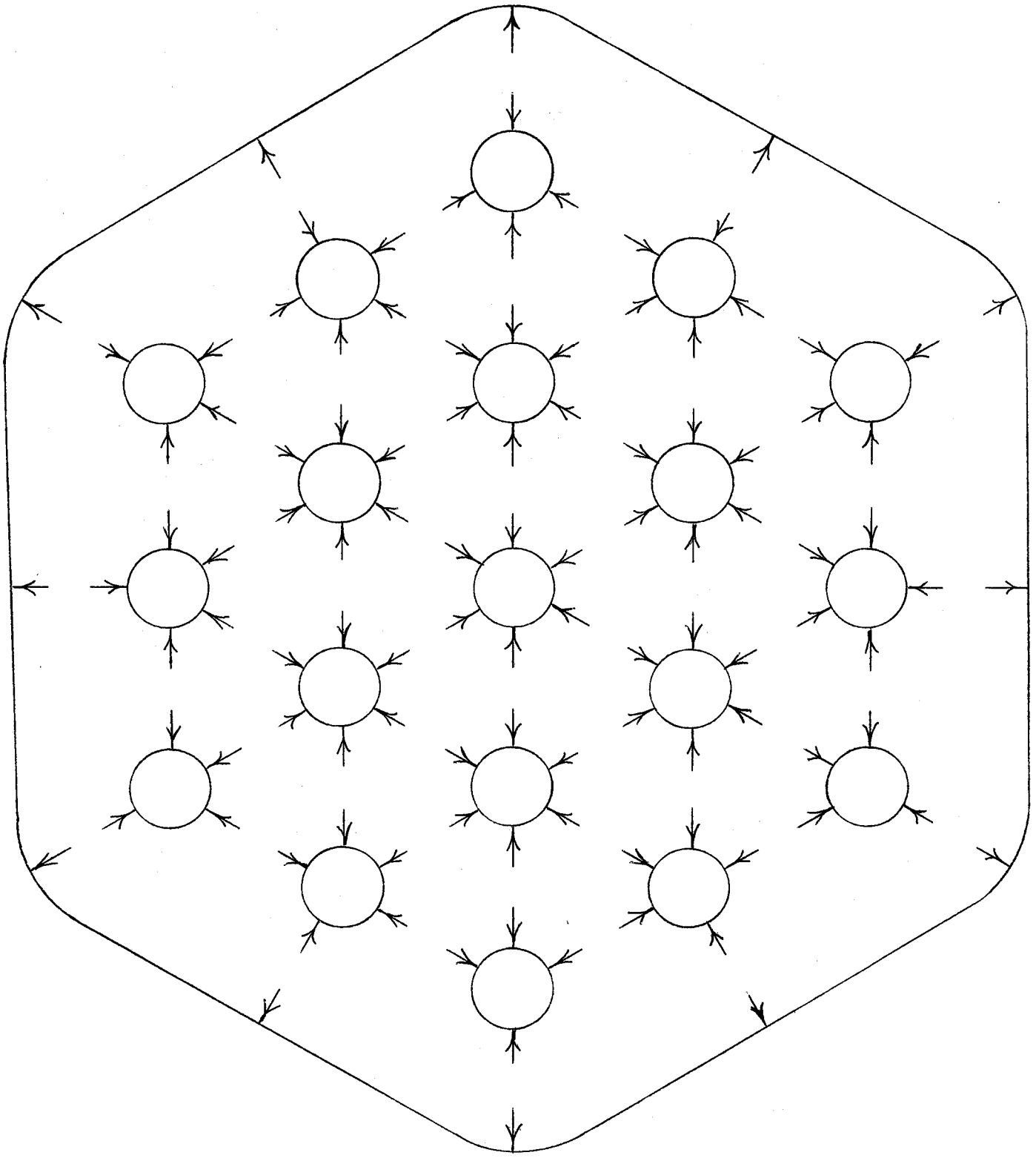


Figure 1. Data Sheet (End Measurements)

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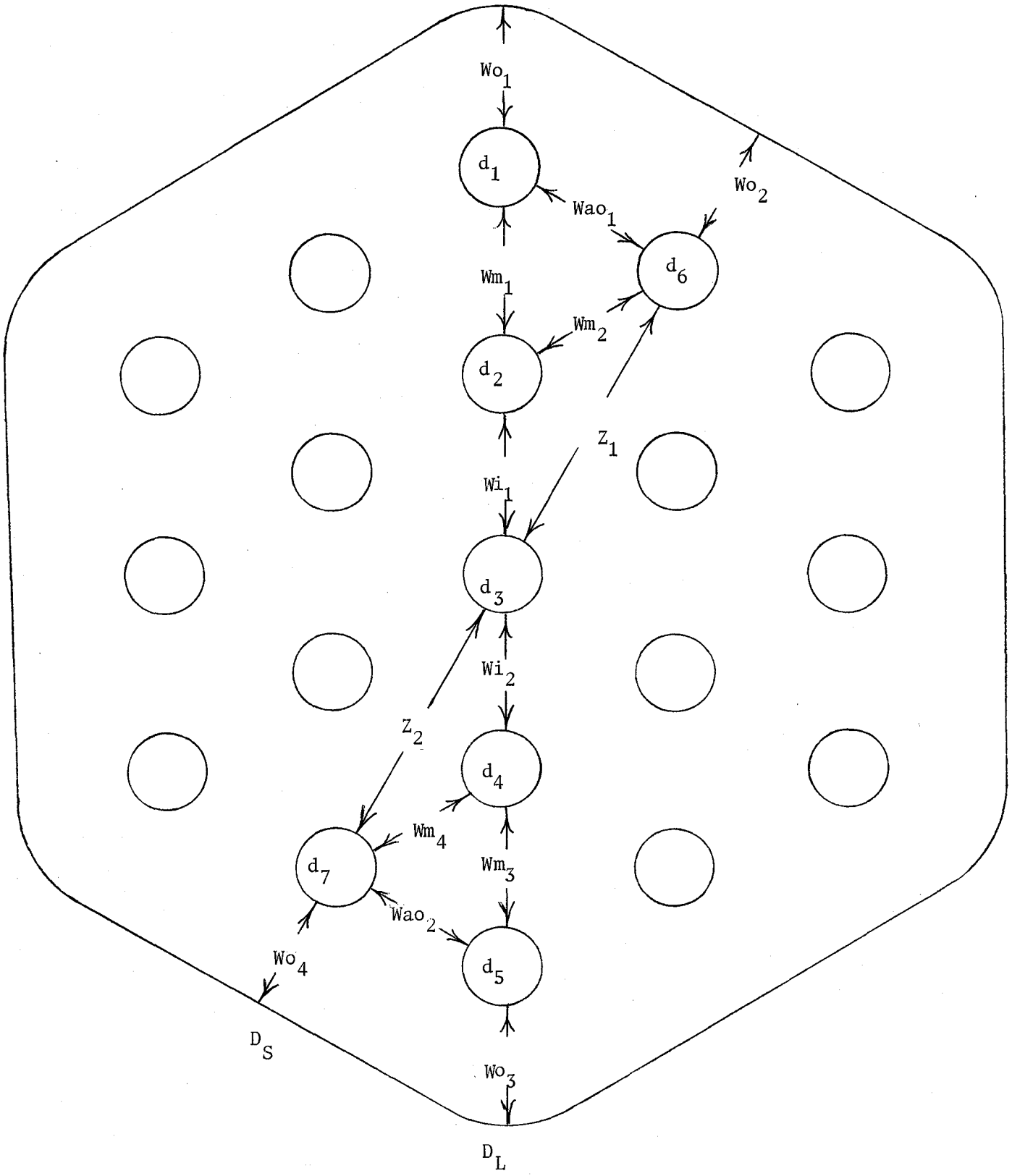


Figure 2. Sketch of 19 Perforated Rounded Hexagonal Configuration (Internal Measurements)

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TABLE II. Proving ground test summary

Test Phase	Temp (°F)	Sample Size		Velocity (fps)		Max Pressure (psi/100)		
		Cal	Test	Average	Std dev Max	Average	IMP	Average plus three std dev
Charge Establishment	70	3	9	4900	-	670	-	-
Uniformity	-35***	5*	10**	-	-	-	-	-
	70	10	10	4900	9.6	670	-	-
	125***	5	10	-	-	-	757	757

* Shall be fired for preproduction only.
** Shall be fired for preproduction and initial production.
***Differential pressure and ignition delay (see 4.5.1.2.4).

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6.4 Surveillance sample. Samples shall be forwarded to Commander, ARDC, US Army Armament, Munitions and Chemical Command, ATTN: DRSMC-LCE-MP (D), Dover, New Jersey 07801 for the propellant surveillance test program in accordance with MIL-STD-652.

6.5 Reference sample. Whenever a propellant lot is selected for reference use, a 5-lb sample shall be forwarded to Commander, ARDC, US Army Armament, Munitions and Chemical Command, ATTN: DRSMC-LCA-G (D), Dover, NJ 07801, for use in evaluating the propellant stockpile.

6.6 Propellant assessment. An additional propellant assessment may be required for a specific production lot of projectiles whose average as fired weight has been found to significantly deviate from the standard projectile weight.

6.7 Calibration cartridges. Because of the sensitivity of the ballistics of the cartridges to the loading procedures, and in order to maintain homogenity, the master calibration or reference lot, as required shall be loaded by the contractor responsible for LAP of this cartridge in production. The propellant assessment cartridges to be tested in accordance with this specification shall also be loaded by the contractor responsible for LAP.

6.8 Same basic granulation. Same basic granulation shall be defined as allowing die changes that result in an average web change of less than 0.0020 inch.

Custodian:
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